

A Certificate-Producing Cascade for Equational Implication: The SAIR EQT2 Stage 2 Solver

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Abstract

The SAIR Mathematics Distillation Challenge on Equational Theories asks a solver to classify whether one magma identity implies another and, for either verdict, to return a certificate accepted by a deterministic Lean 4 judge. We present a single-file solver organized as a cheapest-first cascade. Its false branch combines coefficient tests over structured algebra families, bounded finite-model search, an explicit central-groupoid witness, and several infinite-carrier witnesses. Its true branch is a proof-producing ordered unit superposition procedure with Knuth–Bendix ordering, bidirectional demodulation, indexing, memoised substitution, and anytime size deepening. Search results remain outside the trusted base: successful derivations are replayed as small Lean 4 terms, and countermodels are rechecked by the competition judge.

The frozen solver is a 189,504-byte Python file with SHA-256 `f2392533c9f4c03b...`. In local runs through official judge revision 2848228, it produced accepted certificates for all 1,889 rows of the six public sets with no language-model calls. Separate measurements recorded full agreement on the 800 published Stage 1 evaluation-distribution problems, 100 accepted rows in the canonical Marathon manifest without tokens, and 200 accepted rows in the hosted playground. These are regression and playground measurements, not a leaderboard result and not evidence about a hidden set. All quantitative claims are tied to immutable result ledgers; the paper makes no completeness or comparative-superiority claim.

1 Introduction

An equational theory of magmas has one binary operation and no assumed associativity, identity, or cancellation law. Given equations E_1 and E_2 , the implication problem asks whether every magma satisfying E_1 also satisfies E_2 . The SAIR Mathematics Distillation Challenge, organized by Damek Davis and Terence Tao with the SAIR Foundation, turns that question into a certificate-production task grounded in the Equational Theories Project [3, 12]. A true answer must contain a Lean 4 proof of the universal implication. A false answer must exhibit a magma in which the hypothesis holds and the conclusion fails. The same deterministic judge checks both forms. An accepted certificate solves the row; a plausible label, proof sketch, or partly checked derivation receives no credit [11, 12].

This contract changes the design objective. A classifier may be useful while remaining difficult to audit, whereas a certificate producer must preserve enough structure to reconstruct a kernel-checkable witness. At the same time, the kernel boundary permits aggressive untrusted search. A defect in our term ordering, unifier, model finder, or certificate printer may increase resource use or prevent a solution from being found, but it cannot cause an invalid certificate to be accepted. The competition proxy records success only after the official judge has compiled the submitted term and checked its dependency policy.

Our system implements a deterministic-first cascade in one Python file. The early stages recognize direct implications and search inexpensive countermodel families. A bounded irregular model finder and deeper structured families handle false residuals. An ordered unit-superposition engine handles true residuals and records a replayable derivation rather than only a refutation bit. The same source dispatches to the Solo protocol, in which one process handles one problem, or the Marathon protocol, in which one process shares a global budget among a manifest of problems. Every successful path ends in a Lean 4 certificate.

The central empirical result is deliberately narrower than a completeness claim. The cascade covers every row in the measured public, distribution-drill, and hosted-playground corpora within the recorded budgets. Neither the countermodel families nor the superposition implementation is proved complete, and the deeper stages stop at wall-clock limits. We therefore do not infer performance on the held-back evaluation set. The result is a system description and an auditable regression report, not a theorem about all single-identity implications.

This paper makes four contributions.

- We present a certificate-producing cascade that integrates complementary algebraic model families and proof-producing superposition behind one competition interface.
- We describe certificate encodings that satisfy the judge’s dependency policy, including encodings for finite carriers beyond the direct table helper’s single-digit range and core-only proofs over infinite carriers.
- We analyze two consequential implementation failures: retention of tautologies in the active superposition set and inconsistent handling of the two accepted

spellings of the magma operation. End-to-end certificate runs, rather than label accuracy alone, exposed both failures.

- We provide a hash-bound claim ledger that connects the solver bytes, judge revision, result files, and every evaluation-table row to SHA-256 identifiers.

The rest of the paper follows the execution path. Section 2 gives the single-file architecture. Sections 3 and 4 describe the countermodel and proof branches. Section 5 covers deployment constraints, and Section 7 reports measurements. We then state the reproducibility boundary and place the system relative to equational-theory, superposition, and proof-assistant work.

2 Single-File Solver Architecture

2.1 Contract and trusted boundary

The submission contract requires one `solver.py` file no larger than the competition limit. Solver v2.5 is 189,504 bytes, and therefore remains below that limit while containing all search code and fixed data. In Solo mode the proxy sends one JSON problem and its budget to a fresh process; the solver sends judge or language-model requests back over line-delimited JSON. In Marathon mode the runner supplies a JSONL manifest and an append-only output path. The same source supports both modes, selected by the runner-owned Marathon environment variable [11].

The trusted computing base for a verdict is not the Python implementation. The judge generates the problem statement and the verdict-specific Goal type. A true goal has the form

$$\forall(G : \text{Type}) [\text{Magma } G], \quad E_1(G) \rightarrow E_2(G),$$

whereas a false goal existentially quantifies a carrier and magma for which E_1 holds and E_2 does not. The solver supplies a definition named `submission`; Lean 4 elaboration, kernel checking, and the dependency inspector determine acceptance. Search is therefore an untrusted producer of candidate proof objects.

The interface creates three distinct obligations. First, a candidate must have the right polarity: a universal implication for a true verdict or an existential separating structure for a false verdict. Second, the term must inhabit the judge-generated goal rather than a solver-reconstructed copy that might differ in variable order or parsing. Third, all transitive dependencies must satisfy the proof policy. The judge enforces the first two by type checking `submission : Goal`; its inspection pass enforces the third. The emitted source never uses admitted terms. This separation is why a successful Python-side check is called a candidate throughout the implementation, while “accepted” is reserved for the judge status.

The polarity boundary also determines failure recovery. The solver does not cache an unverified Boolean verdict and

ask later stages merely to justify it. Each stage returns a self-contained candidate of one polarity, submits that candidate, and continues if checking fails. For example, an arithmetic countermodel whose certificate exceeds an instance budget does not bias the proof branch toward false; it is discarded as an unavailable witness. A superposition trace that cannot be replayed is likewise not evidence for a true output. This discipline prevents a search heuristic from escaping into the externally visible result.

At both protocol entry points we normalize the input operation symbol. The official problem format permits either an asterisk or a diamond, but the solver’s term parser has one internal token. Mapping the asterisk to $\diamond$ before parsing preserves the semantics and is operationally necessary; Section 5.2 records the failure that revealed it.

2.2 Cheapest-first cascade

The Solo cascade is ordered by expected cost and by which branch benefits from remaining time.

Stage A: shallow countermodels. The solver first tries tiny brute-force tables, scalar linear and affine operations, vector-linear families, low-degree polynomial operations, and recognized infinite-carrier Austin models [1]. The expanded structured families add about 0.8 seconds on a true input in the recorded development measurement. No irregular finite-model search is performed yet.

Stage B: direct true certificates. Syntactic singleton collapse and substitution-instance checks produce direct proofs. A specialized singleton-forced prover then receives an eight-second cap, although applicable cases normally close during its initial search. The general superposition engine follows with a quick budget $\min(6, \max(1, T/600))$ seconds for a process allowance T ; this is six seconds under the recorded Solo allowance. Most provable public instances terminate within this cap.

Stage C: bounded irregular models. A finite-model search over carriers four through ten receives eight seconds. It propagates the universal instances of the hypothesis and Latin-square constraints rather than enumerating complete tables.

Stage D: deep false search. Only residuals reach heavier polynomial and vector families, capped at sixty seconds, followed by another finite-model pass capped at one hundred twenty seconds. Deep false search precedes the deepest proof pass so that a difficult false row does not consume most of its allowance in proof saturation.

Stage E: anytime proof search. The ordered-superposition engine, in the tradition of completion-based equational provers [2, 8], receives $\min(600, \max(20, T/6))$ seconds, which is six hundred seconds under the Solo allowance. The engine repeatedly raises term-size and variable caps. If no direct refutation is returned, a small pool of derived equalities can

be emitted as exact lemmas followed by a kernel-checked closing tactic. A language-model feedback loop follows the deterministic cascade, but none of the accepted rows in the archived regression invoked it.

Every stage independently verifies its candidate before the solver stops. A rejected certificate proceeds to the next strategy; a successful search without a valid emitted term is not counted as a solution. This control flow distinguishes search coverage from certificate robustness.

2.3 Representation and dispatch

Both branches share a small non-associative term representation. Parsing produces a binary tree with variables at leaves; no reassociation or commutative normalization is performed. Alpha-renaming maps variables to a canonical local order for cache keys and family matching, but emitted proofs retain a map back to the judge's binder order. Operation count, variable support, depth, and tree size are computed once and reused by scheduling, countermodel probes, and proof search.

Dispatch relies on positive recognizers. A direct proof stage returns only after constructing a substitution or equality chain. An Austin or central-groupoid stage returns only when the hypothesis matches its stored schema and a concrete evaluation separates the goal. A family that is merely likely from equation identifiers is not selected: identifiers are useful for diagnostics, but the solver matches parsed laws. This property supports independent replay and handles inputs whose labels or variable names differ from those in the public files.

Candidate validation is stratified. Low-cost Python evaluation removes malformed tables, wrong witnesses, and printer bugs before a judge call. Judge feedback then closes the semantic and dependency obligations. The development harness records both outcomes: a "found" countermodel rejected by Lean 4 indicates an emitter defect or an encoding mismatch, whereas the absence of a Python witness within a deadline indicates a coverage limitation. This distinction informed the large-carrier encoding correction in Section 3.5.

2.4 Solo and Marathon

Solo gives each row a fresh process and a fixed allowance. Marathon instead requires triage across a whole manifest. The batch path stably sorts problems by a structural cost derived from operation occurrences, variable count, and equation length. It then runs two passes. The first gives every row the shallow stages before any residual receives deep work. The second revisits only residuals. A fair cap is recomputed from remaining wall-clock time divided by remaining rows and clamped to the documented per-row interval. At most fifty-five percent of a residual's current slice goes to the false branch; the proof branch receives the rest.

Marathon outputs constitute durable state. Each answer is one JSON line, flushed and synchronized before the next

row. If the runner terminates the process, all complete earlier lines remain scoreable. A process-level timer bounds legacy search loops that predate the batch driver. The batch path issues no speculative language-model calls and leaves a residual unanswered rather than consuming tokens without producing a certificate.

2.5 Cascade invariants and failure isolation

Stage ordering does not assert that either polarity is more likely. Instead, it minimizes expected certificate cost while reserving time for both branches. A structured false witness is attractive early because checking a closed operation is usually cheaper than saturation. Direct true patterns are equally attractive because they avoid both table construction and general proof search. Measurements showed that most provable rows close near the start of saturation, which motivates placing the quick G3 pass before irregular model search. Deep false search precedes deep G3 only after every low-cost recognizer has failed.

The implementation maintains a cascade invariant: before entering a stage, the solver has no accepted certificate, and every earlier candidate has either been absent, failed local construction, or been rejected by the judge. It does not maintain the stronger and generally invalid invariant that earlier stages have ruled out their polarity. Failure to find a finite model does not support truth; failure to derive the goal does not support falsity. Each stage can therefore be analyzed and tested as a partial certificate producer, and the composition is sound whenever the judge is sound, regardless of the producers' coverage.

Exceptions are contained at strategy boundaries in the Marathon path so that a malformed residual cannot discard previously completed rows. Solo has a fresh process per row and can rely on the runner to record a crash. Within a strategy, deadlines are checked at natural allocation points: carrier changes, model-search decisions, deepening rounds, and given-clause iterations. The Marathon alarm provides a final outer bound for code paths whose internal checks are too sparse. Timeout returns "no candidate" rather than a verdict.

Caches follow the same lifetime discipline. Parser and static-law data live for the process. A superposition run owns its unifier, normalization, term, and clause caches, which are cleared before another problem. Marathon could in principle share more proof state across rows, but the current implementation does not rely on cross-problem lemmas. This keeps Solo and Marathon certificate semantics aligned and limits memory growth under a long manifest.

The fallback model follows the complete deterministic cascade in Solo and is not invoked in the archived accepted rows. Its proposals still pass through the same printer and judge, so adding the fallback does not enlarge the trusted base. In Marathon it is disabled to preserve the global token

budget; the result is a partial manifest containing only verified deterministic answers. These are scheduling choices, not claims about the intrinsic capabilities of the configured model.

3 Countermodel Construction

For a false implication the solver must find an operation satisfying all assignments of the hypothesis and at least one assignment falsifying the goal. It evaluates candidate families in Python, retains a concrete witness, and emits a certificate in which the judge repeats the decisive checks.

3.1 Coefficient matching over structured families

Represent a magma term by the formal coefficients induced by a candidate operation. For

$$x \diamond y = ax + by \pmod{n},$$

each leaf contributes a coefficient obtained by multiplying edge labels along its path. Equality of two terms for all assignments is therefore equality of their coefficient vectors modulo n . This gives an exact, inexpensive test for whether the hypothesis is an identity of the candidate magma. The same representation quickly locates an assignment on which the goal coefficients differ. The scan includes composite moduli and its shallow range extends to the largest carrier admitted by the certificate-instance guard.

Affine operations add a constant term, and matrix-linear operations replace the scalar coefficients by matrices over small finite fields. The latter searches include two-dimensional vectors over the three-element field and three-dimensional vectors over the two-element field. Polynomial candidates are evaluated on full assignments when the carrier and hypothesis arity make that feasible. The deep pass adds a quadratic grid, a two-dimensional family over the five-element field, and sampled polynomials of degree at most three. An instance-count guard prevents a mathematically compact operation from producing a certificate whose exhaustive hypothesis check exceeds the judge budget.

These families provide both search efficiency and structured certificates. They find many witnesses faster than a generic table search while preserving semantic structure in the emitted certificate. For example, a linear countermodel can be represented by its arithmetic operation rather than by a large opaque table. The judge still evaluates the hypothesis and the failing goal instance, so coefficient matching remains an untrusted search optimization rather than a proof rule.

Coefficient evaluation is implemented recursively. A variable leaf maps to its basis vector. At an internal node, the coefficient vector is $a\vec{u} + b\vec{v}$ for child vectors $\vec{u}$ and $\vec{v}$; affine families carry one additional constant coordinate. Thus the universal hypothesis test does not enumerate assignments. If the two sides have different vectors, the candidate is immediately rejected as a model of the hypothesis. If the hypothesis

vectors agree and the goal vectors differ, a separating assignment can be sought in the small module. Full evaluation is still applied before emission, which protects the certificate path from an error in this symbolic shortcut.

Vector-linear families use exactly the same recursion with matrices acting on coordinates. They are useful because scalar coefficient coincidence may hold in every small modulus even when non-commuting linear actions distinguish the goal. Polynomial operations abandon the linear invariant and instead use compiled term evaluators over bounded carriers. The family ordering therefore progresses from symbolic coefficient equality to increasingly expensive full evaluation, while presenting a uniform result to the certificate printer: a carrier, a total binary operation, and a goal witness.

3.2 Irregular finite models

Some countermodels have no low-degree algebraic description. The finite-model stage treats the Cayley table as a constraint problem. For a proposed carrier size, table cells begin unassigned. Ground instances of the hypothesis watch the cells on which their two terms depend. Assigning a cell may simplify a term, force another cell, or violate an instance. The search uses a minimum-remaining-values choice among table positions and abandons partial tables as soon as a hypothesis instance becomes unequal.

Many identities imply that rows, columns, or both must be permutations. A law in which a variable occurs once on one side and appears under the operation in a particular argument position can expose such a constraint. The solver infers the applicable Latin flags, rejects duplicate row or column values, and forces naked singletons. For one residual requiring an eight-element quasigroup, this propagation changed the outcome from no witness within a one-hundred-twenty-second attempt to a witness in 0.3 seconds. The production shallow search remains restricted to carriers four through ten and an eight-second cap; the deep pass revisits the same carrier interval with a longer cap.

Before emission, the completed table is checked by a separate evaluator over all hypothesis assignments and a stored goal-falsifying assignment. The Lean 4 certificate then reconstructs the table operation and invokes the judge's finite-decision tactic. The duplicated checks reduce wasted judge calls, while only the latter determines acceptance.

The partial-table solver maintains two invariants. Every assigned cell agrees with all hypothesis instances that have become ground, and every inferred Latin constraint agrees with the assigned portion of its row or column. A watch list connects a cell to the term instances whose evaluation may advance when that cell is filled. Propagation reaches a fixed point before the next decision. At a complete table, the first invariant implies that exhaustive reevaluation of the hypothesis should succeed; the separate final evaluator checks this implication and also searches the goal assignments. Search backtracking may be unsound or incomplete as an algorithm

without threatening an accepted answer, because the table is subsequently evaluated from scratch and then checked once more in Lean 4.

Carrier growth is likewise heuristic. The shallow and deep passes enumerate the documented carrier interval, but the solver does not infer that absence of a model there implies truth. It passes the row to the next proof or model family. This behavior implements the paper's no-completeness boundary.

3.3 Central groupoids and the E168 residual family

The evaluation-distribution drill identified a family of twelve residuals whose hypothesis is the central-groupoid law. Central groupoids have a well-known combinatorial structure studied by Knuth [7]. The natural constructions tried by the structured stages satisfy both the hypothesis and the relevant E168 conclusions, so they do not separate these equations. Increasing the generic finder budget did not reliably recover the missing witnesses.

The solver therefore includes one explicit non-natural central groupoid of order nine. The table was checked against each relevant hypothesis and goal, then stored once as a fixed witness family. When the hypothesis matches up to variable renaming and a goal is separated by the table, the solver emits the ordinary finite-table certificate. This stage is not a classification of central groupoids and makes no claim about minimal order. It provides a specific, auditable witness for a recurring algebraic family: the official runner accepted the emitted certificates for the twelve drill residuals.

3.4 Infinite carriers for Austin pairs

Finite satisfiability can conceal false general implications. Several pairs catalogued in the Equational Theories Project are true over the finite magmas searched by standard methods but false in general. For recognized Austin-pair hypotheses, the solver selects a fixed operation on $\mathbb{N}$, or its opposite, and a small tuple falsifying the goal. The certificate proves the hypothesis through a fixed sequence of elementary lemmas and closes the negated goal with the concrete tuple.

This is qualitatively different from finite decision. There is no exhaustive tactic over $\mathbb{N}$, so the complete mathematical argument must be expressed in the certificate. The original certificates imported a broad tactic module; Section 5.1 describes why those imports became an operational problem and how positive-form replay reduced their compile time.

3.5 Certificate shapes and the allow-list

For carriers through ten, `finOpTable` embeds a JSON-like Cayley table and `decideFin!` checks the finite existential goal. The helper extracts individual decimal characters, so a table entry with multiple digits is not a faithful encoding. Larger carrier tables therefore cannot use that direct shape.

The alternative is a definition such as `submission.op` whose body computes a linear, affine, polynomial, or packed-table lookup operation. The judge checks its allow-list on

the direct constants of the submitted term; helpers inside the submission namespace are unrestricted, which the organizers document as intended for auxiliary definitions. The final theorem refers directly to the namespaced operation and core arithmetic. This route was accepted by the official judge for the recorded larger carrier sizes and avoids relying on the table parser.

This distinction affects dependency enforcement but does not circumvent it. The helper definition is part of the submitted source, is compiled by Lean 4, and is traversed by the kernel when its theorem is checked. The allow-list controls dependencies, not the truth of definitions. As specified by the documented proof interface, submission-local auxiliary definitions provide the intended mechanism for packaging operations and lemmas that are not global library dependencies.

4 Proof-Producing Ordered Superposition

The true branch operates in unit equational logic. It assumes one universally quantified identity and attempts to derive the goal identity. The engine is inspired by ordered superposition and unfailing completion as implemented in first-order provers, but is specialized to one binary symbol, unit clauses, and the need to reconstruct a small positive Lean 4 proof.

4.1 Terms, ordering, and inference

Terms are immutable trees over variables and the binary magma symbol. A Knuth-Bendix ordering with unit symbol weights determines maximal equation sides and orients usable rewrite rules. Superposition overlaps an oriented side with a non-variable subterm of another maximal side, computes a most general unifier, and constructs the resulting equality. Ordering checks are repeated after unification, because substitution can change comparisons. Equations that cannot be oriented remain available for symmetric inferences.

The goal is represented internally as a negative unit clause $u \neq v$ over fresh constants. Active equalities can superpose into either side; ordinary demodulation simplifies the clause. If its sides unify, the negative clause is refuted. This internal negative representation is convenient for search, but the emitted proof does not ask Lean 4 to validate a refutation calculus. It instead reconstructs a positive equality chain ending at the original goal.

Forward demodulation normalizes each generated equality by current rules. Backward demodulation revisits active equations when a new rule can simplify them. Symmetric alpha-normalized keys remove variants. A given-clause loop interleaves a weight heap with an age queue; later deepening rounds alter the variable penalty and age ratio so that a proof blocked by one selection policy is not permanently starved.

4.2 The tautology-deletion failure

An early version of the prover did not resolve one public residual, `hard3_0314`. The official-runner attempt found no proof after 810 seconds even though an external derivation suggested that the necessary clauses were within the configured size bound. Instrumentation indicated that throughput was not the primary cause. When demodulation transformed an equation into $s = s$, the step constructor returned no proof node, and the caller retained the partially rewritten input equation. Backward simplification had the same behavior.

Consequently, many small instances of the hypothesis remained live. Their weights made them attractive to given-clause selection, so they repeatedly displaced clauses on the useful derivation. The correct simplification rule is deletion: a demodulated tautology contributes nothing to saturation. The fix retires the active clause before testing the rewritten result and drops a new clause whose sides become equal.

With that change, the prover found the residual proof in 0.6 seconds in its direct measurement, and the official-runner path produced an accepted certificate in 14.6 seconds of wall-clock time. These measurements illustrate the coupling between simplification and selection rather than establishing a general speed ratio. Retaining a tautology as a live clause changed the effective search strategy enough to prevent the final public proof from being found within the earlier attempt.

4.3 Indexing and allocation control

Demodulation candidates are retrieved through a discrimination tree keyed by a term's linear skeleton. The index returns rules whose left sides may match a subterm; matching and the ordering check filter false positives. Normal forms are cached and versioned by the rewrite-rule set. Term sizes, variable multisets, preorder traversals, and eligible positions are cached on immutable equations.

The unifier returns a triangular substitution rather than eagerly applying it to every binding. A memoised substitution routine resolves chains while sharing unchanged subtrees. Before constructing a superposition result, the engine estimates instantiated size from static term sizes and per-variable counts. Overlaps that cannot fit the current cap are rejected before allocating their terms. Unifier results are cached by the renamed overlap pair. These changes preserve the inference trace at fixed bounds while reducing the work per surviving clause.

The passive set stores recipes—source equation, target equation, direction, and overlap position—rather than full instantiated proof objects. The selected recipe is materialized when it becomes given. Under the hosted memory limit, this representation permits a large passive frontier without duplicating every term and ancestry chain. Caches and negative-goal storage are bounded and cleared between prover invocations.

4.4 Anytime deepening

A fixed size cap is sensitive to the selected bound: saturation at a small cap can terminate with time remaining even when one slightly larger intermediate would finish the proof. The deep pass therefore runs a schedule of increasing term-size and variable caps. If a round saturates, its unused time moves forward. If the final configured round saturates, caps continue to grow within the overall deadline. The last strategies reduce the variable penalty and increase age pressure, approximating a different prover configuration without maintaining another implementation.

The procedure remains budget-bounded and machine-load sensitive. It is not a new completeness result for ordered superposition, nor a proof that the chosen ordering is fair under every cap. The evaluation in Section 7 measures this implementation on fixed corpora.

4.5 Replay into the Lean 4 kernel

Every retained equation stores a derivation recipe. Replay recursively reconstructs source equations, instantiates the universally quantified hypothesis, and uses congruence to place an equality inside the overlap context. A superposition step becomes a have binding assembled from `congrArg`, symmetry, and transitivity. Goal simplifications become another exact equality chain. The final proof is a sequence of `have/rw/exact`-style core terms in positive form.

Positive replay isolates search from trust. The Python engine can orient an equation incorrectly, use a stale cache, or reconstruct the wrong path; any of these produces a term that fails to elaborate or has the wrong type. The judge then rejects it, and the solver may retry robust emission or continue to another stage. A prover defect can therefore cause a missed proof or a slower run, but cannot cause a wrong acceptance unless there is also a defect below the Lean 4 kernel boundary.

Replay granularity and certificate size impose competing requirements. Emitting every normalization micro-step is easy to audit but repeats contexts and substitutions. Compressing an entire derivation into a large tactic call reduces source size but reintroduces a dependency on tactic behavior and library imports. The solver uses one named equality per selected inference, with normalization represented as explicit transitivity where needed. Shared ancestors are emitted once in topological order. This keeps proof text close to the saturation trace while allowing Lean 4 to elaborate each local equality without reconstructing the global search.

Robust re-emission addresses elaboration rather than logical inference. The first printer uses direct exact terms. If the judge rejects that source, the solver can wrap individual equality steps in a small sequence of core closing alternatives and resubmit the same derivation; it does not rerun superposition or change the claimed theorem. If search exhausts its deadline without a refutation, the lemma-pool path emits

a bounded selection of the smallest already proved consequences. Any final tactic must still close the judge goal from exact lemmas. Failure to do so leaves the row unanswered.

5 Engineering for the Hosted Environment

5.1 Core-only certificates

The public harness used Lean 4 4.30.0-rc2, while the announced hosted verifier used Lean 4 4.32.0 with the corresponding Mathlib release. Most emitted certificates already imported only judge modules and core definitions. The exception was the infinite-carrier Austin family, whose tactic import made compilation sensitive to the larger library environment. In a compatibility run, one such certificate required about 330 seconds of compilation.

Rather than increasing the solver-side timeout, we rewrote the Austin certificate in positive form, replaced the broad tactic dependency with explicit core lemmas, and re-emitted the witness. Its Lean 4 4.32 compilation then completed within 3.1 seconds. A stratified corpus of 120 accepted certificates, covering every emission family and all fourteen former v1 residuals, compiled 119 of 120 under the exact announced toolchain in an external harness with a 120-second per-phase limit; the one non-pass was the pre-rewrite Austin witness itself, which passes under a 300-second per-phase configuration matching the judge (phases of 184 and 145 seconds). The corpus therefore characterizes the migration-era certificate set rather than the frozen artifact's current output: the frozen solver's own certificate for the same problem imports only the judge module and is accepted in under five seconds. The judge support modules compiled unchanged. That corpus run executed outside the repository and is archived as an operator attestation (`results/lean432_corpus/`). When the organizers subsequently upgraded the judge to Lean 4 4.33.1 after the kernel-soundness review, the frozen artifact was revalidated in full under the upgraded judge — official harness, all six public sets, the distribution drill, and the Marathon manifest — with the resulting ledgers committed and hash-bound (`results/lean4331_revalidation/`). Neither measurement substitutes for hosted scoring.

5.2 Input encoding

The official judge normalizes the asterisk and diamond spellings when it builds the Lean 4 problem. The runner, however, passes the original problem text to the contestant process. An earlier solver normalized only along one internal path. When the Stage 1 evaluation splits were supplied in their published asterisk encoding, the solver crashed before certificate search and recorded zero accepted rows out of 800.

Version v2.4 introduced, and v2.5 retains, normalization of both equations at both the Solo and Marathon intakes. The same drill then recorded 800 accepted certificates with full agreement against the published answers. Thus, verified output does not address failures in an unverified parser;

normalization must occur at the protocol boundary, before any dispatch or caching.

5.3 Resource and filesystem constraints

The hosted-style sandbox has a read-only submission mount, a bounded process count, bounded memory, no direct network, and a small writable temporary filesystem. The solver consequently uses the Python standard library, keeps all static data in the single source file, does not spawn external provers, and does not assume repository access. Bounded passive queues and caches keep the superposition engine below the sandbox memory ceiling observed in long false saturations. The process-level timer also prevents a deeply nested legacy loop from overrunning its Marathon slice.

Marathon's output file is the only durable mutable state supplied by the runner. Per-row flush and synchronization are therefore required for correctness. An initial local fixture run also showed that the official checkout's default artifact directory could be unwritable in a managed workspace; redirecting judge artifacts to temporary storage resolved the environmental failure without changing judge sources.

6 End-to-End Certificate Production

Search and replay together form a single transaction. This section follows representative false and true candidates from a parsed equation pair to the theorem checked by the judge. The examples show certificate shape rather than a particular benchmark row; the concrete generated sources remain in the archived result ledgers.

6.1 Finite countermodel transaction

Suppose coefficient matching or table search returns a carrier $\text{Fin } n$, an operation m , and an assignment ρ that separates the goal. Before printing Lean 4, the solver performs three checks. It confirms that every table entry or arithmetic result belongs to the carrier. It enumerates every assignment required by the hypothesis and checks equality of the evaluated sides. Finally, it reevaluates the two goal terms at ρ and checks that their values differ. A failure at any point discards the candidate without a judge call.

For a small table the emitted proof has the following schematic form:

```
def submission : Goal := by
  let m : Magma (Fin n) :=
    { op := finOpTable "<table>" }
  refine <Fin n, m, ?_>
  decideFin!
```

The angle-bracket line is printed with Lean 4's tuple notation in the actual source. The judge's decision tactic expands the finite universal hypothesis and the existential goal witness under the supplied operation. Thus the stored Python assignment is useful for prechecking, but the accepted theorem does not rely on the Python evaluator.

For a larger structured carrier, the transaction changes only in how m is defined. A helper inside the submission namespace computes the operation using Nat and Fin primitives. The theorem refers to that helper when constructing the magma, and finite decision closes the same logical goal. Packed-table witnesses use a natural-number payload and an indexing expression in the helper body. Arithmetic families print their closed form directly. Both routes keep multi-digit entries away from finOpTable while leaving the operation transparent to the kernel.

The certificate has two independent failure modes. The operation may not be a countermodel, in which case finite decision fails with an ordinary proof error. Or the source may use a dependency outside the policy, in which case inspection classifies it as incomplete. Development probes must run through the complete official policy: an earlier permissive harness omitted the declaration restriction and made a rejected arithmetic shape appear usable. Only the official runner result was retained as evidence.

6.2 Infinite countermodel transaction

An infinite witness cannot delegate universal checking to finite reflection. The Austin path first matches the parsed hypothesis against a stored law up to renaming. It selects an operation $f : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}$, instantiates a fixed proof that f satisfies the law, and evaluates candidate tuples until the goal sides differ. Emission packages $\mathbb{N}$, the magma instance, the hypothesis theorem, and a function that turns any assumed goal equality at the chosen tuple into a contradiction.

The hypothesis theorem is the expensive part of the source. Its core-only version is a sequence of named local facts about the operation followed by the target identity. The goal refutation is intentionally concrete. It applies the universally quantified conclusion to the stored natural-number tuple and uses computation to show that the resulting equality is impossible. This division makes the reusable part of the mathematical construction visible while keeping each problem-specific separation small.

As with finite models, recognizing the hypothesis is not itself trusted. A renaming bug would instantiate a lemma at the wrong law, and Lean 4 would reject the resulting type. A wrong tuple would fail to close the negation. The kernel therefore checks both the reusable model theorem and its application to the current judge-generated equations.

6.3 Superposition replay transaction

A successful G3 search returns a refuted negative goal plus a directed acyclic graph of selected inference recipes. Replay begins from the nodes reachable from the refutation and topologically orders them; irrelevant saturation clauses are omitted. Hypothesis leaves become applications of the input identity with a substitution for its universally quantified variables. Derived leaves created by symmetry or reflexivity become the corresponding core constructors.

For an overlap, replay first obtains the equality used as a rewrite. It lifts that equality through the surrounding term context with congrArg , choosing symmetry if the search used the reverse direction. It then composes the lifted equality with the instantiated target equation by transitivity. The generated source consequently resembles

```
have eA (...) : lhsA = rhsA := by exact h ...
have eB (...) : lhsB = rhsB := by
  have lifted := congrArg C (eA ...)
  exact lifted.trans ...
...
exact finalEquality
```

where actual variable binders, contexts, and equation sides are printed from the replay graph. No saturation data structure or KBO comparison is imported into Lean 4. The certificate states only the equational consequences needed for the final chain.

This proof shape also supports robust recovery. If an exact term is difficult for elaboration because an implicit argument is ambiguous, the robust printer adds local type information or a small closing alternative around that step. It never replaces a missing inference with an axiom. Because reachable nodes are emitted once, a clause used by several descendants is shared by name rather than duplicating its derivation.

6.4 Acceptance as the commit point

In Solo mode the solver sends the complete source in a judge request. The proxy writes the judge-controlled problem module, compiles the candidate, runs dependency inspection, and returns one of the documented statuses. Only the accepted status commits a solution row. All other statuses preserve enough error text for robust re-emission or later fallback, subject to the remaining wall-clock budget.

Marathon cannot make interactive judge calls. Its commit point is therefore split: the solver durably appends a complete candidate row, and the runner checks the last complete candidate for each identifier after process exit. The append-and-synchronize protocol ensures that termination between problems cannot corrupt already emitted source. This difference changes scheduling but not trust: the same verifier and proof policy decide both tracks.

7 Evaluation

7.1 Method and provenance discipline

Unless marked hosted, measurements in this section are local runs of the official runner and deterministic Lean 4 judge. The main regression used official repository revision 2848228ff490...; the harness completed without errors on the operator host. The tested solver is v2.5, SHA-256 f2392533c9f4..., with byte count given in the abstract. Each result file is bound by the repository's PROVENANCE.json. The short hashes in the tables are prefixes of the full SHA-256 values in that file.

The reported wall-clock time is the sum of per-row runner measurements, not elapsed parallel makespan and not a hardware-independent cost model. Solver stages with deadlines may take different paths under heavier load. “LLM” counts language-model calls attributed by the runner; a zero means the deterministic cascade produced the accepted certificate.

7.2 Six public sets

Table 1 reports the final six-set regression. All rows were accepted, every accepted row had one judge call, and no row used the fallback model. The sum of the set sizes is 1,889. The earlier frozen v1 solver accepted 1,875 of those rows; v2 adds the structured false stages and G3 proof engine described above. We report v1 only as development context, not as a comparison against another entrant.

The totals establish regression coverage of these public artifacts only. The private evaluation set is separate, and public-set tuning is possible. We do not treat Table 1 as a hidden-set estimate.

7.3 Component-level interpretation

The superseded v1 baseline left fourteen public residuals: eleven true implications without a deterministic proof inside its budgets and three false implications without a model in its available families and carriers. The v2 development was organized around those failure classes. G3 plus the tautology-deletion correction supplies certificates for the true residuals; the large-carrier, Latin-propagating, structured, and infinite-carrier paths supply witnesses for the false residuals. Replaying the residual set alone records fourteen accepted certificates, whereas the six complete sets provide a regression check that stage ordering and printers preserve the previously accepted cases.

This history is diagnostic, not a controlled ablation study. Several changes were introduced together, and a row may now be solvable by more than one stage. The paper therefore attributes mechanisms only where the engineering notes contain a direct replay, such as `hard3_0314` for tautology deletion and `hard1_0062` for Latin propagation. It does not assign an aggregate percentage gain to individual components.

The order of the cascade also affects the observed wall totals. A true row may incur the shallow false-family cost before receiving a quick proof; a false row may incur direct proof checks before finite-model search. These costs are intentional because the early stages reduce resource use across both branches: inexpensive structured countermodels avoid proof saturation, while direct proof patterns avoid table search. The six-set totals measure the integrated policy, not isolated engine speed.

7.4 Distribution drill, Marathon, and hosted playground

The official-distribution drill uses the four published Stage 1 evaluation splits that were named as Stage 2 scoring categories. Each split contains published ground truth and uses the asterisk input encoding. Table 2 reports full local judge acceptance and full agreement with those labels. Stage 2 does not reuse these problems, so the drill measures format and distribution readiness rather than evaluation transfer.

The canonical Marathon measurement used the official batch runner and its `normal_100` manifest. All rows were accepted without tokens. The hosted-playground row is different in kind: it was run through the official playground interface on the hosted judge and is recorded as an external operator attestation, because the playground emits no repository-committable ledger. Across the four evaluation categories the attested runs record acceptance of every attempted problem with no rejected or errored rows and no language-model calls. It is explicitly not an official leaderboard score.

The hosted record further reports means by verdict: 2.69 seconds for false rows and 7.08 seconds for true rows, with maxima 6.5 and 23.8 seconds, respectively. Those values include hosted orchestration and certificate checking as exposed by the playground, and should not be compared directly to the local prover-only benchmark below.

7.5 Focused prover measurement

To test the G3 search independently of the full cascade, development used a corpus of 254 ETP pairs selected because an existing Vampire derivation was available and the earlier prover found them difficult. Under a thirty-second per-pair prover cap, the optimized engine found all 254 derivations; maximum prover time was 2.7 seconds and the sum was sixteen seconds. On all 819 public true problems under the quick cap, the prover found all proofs with median one millisecond and maximum 1.2 seconds. These are prover-only measurements on the development machine, not judge wall-clock time and not a comparison claiming superiority over Vampire. Vampire served as a source of hard instances and derivation guidance.

These focused results are recorded in `docs/TRUE_SIDE_G3_PROVER.md`; they have no standalone hash-bound runner ledger and are therefore not included in the ledger tables.

7.6 Bounded language-model measurement

Before the v2 deterministic extensions, a fixed-configuration run exercised the organizer-pinned `gpt-oss-120b` fallback on six unsolved `sample_20` residuals. It solved zero of the six. Pilot and canonical runs also produced different first attempts despite the advertised zero temperature and fixed seed. The canonical run cost \$0.1350 in total, measured from provider credits-balance deltas rather than runner logs —

Table 1. Local official-runner regression for solver v2.5. “Wall” is the sum of per-row wall-clock seconds. Each row names its immutable ledger and SHA-256 prefix.

Set	Accepted	LLM	Judge	Wall (s)	Ledger provenance
sample_20	20/20	0	20	67.53	results/v2_sample_20_official_2848228.json b17eaff452d2...
sample_200	200/200	0	200	964.12	results/v2_sample_200_official_2848228.json dcf830de6199...
hard1	69/69	0	69	559.81	results/v2_hard1_official_2848228.json 7724566d1ceb...
hard2	200/200	0	200	1250.67	results/v2_hard2_official_2848228.json 3e68548945d4...
hard3	400/400	0	400	2857.38	results/v2_hard3_official_2848228.json c9f98750e018...
normal	1000/1000	0	1000	4128.3	results/v2_normal_official_2848228.json 27c756482d15...

Table 2. Additional measurements. Local drill and Marathon rows name their ledger files and SHA-256 prefixes. The hosted row is an operator attestation recorded in the hosted_playground_measurement entry of PROVENANCE.json; the playground interface did not emit a repository ledger file.

Measurement	Accepted	LLM/tokens	Wall statistic	Ledger provenance
evaluation_order5	200/200	0 calls	sum 583.29 s	results/official_distribution_drill/evaluation_order5_results.json 422ecdf5a856...
evaluation_extra_hard	200/200	0 calls	sum 1735.26 s	results/official_distribution_drill/evaluation_extra_hard_results.json 99d16aec72f...
evaluation_hard	200/200	0 calls	sum 889.03 s	results/official_distribution_drill/evaluation_hard_results.json 4505248a3531...
evaluation_normal	200/200	0 calls	sum 2288.06 s	results/official_distribution_drill/evaluation_normal_results.json 2377747990d8...
Marathon normal_100	100/100	0 tokens	full default budget	results/marathon/normal_100_summary.json 28fc79269ede...
Hosted playground (evaluation_normal)	200/200	0 calls	mean 4.89 s/problem	PROVENANCE.json:hosted_playground_measurement artifact 7916cbc2...

about \$0.023 per residual item under the sixteen-round cap. Per-call latency and token telemetry are recorded as unavailable in the run notes, so no per-call figures are reported here. We report this bounded negative measurement of one model, provider route, solver version, prompt, and residual set. It does not support a general statement that language models are ineffective for theorem proving or for this competition.

The executed pre-fix solver is bound in docs/LIVE_RUN_NOTES.md by SHA-256 prefix 22a0bc846288...; the canonical row data are in results/sample_20_live.json. Because no ledger-file hash for this historical run appears in the claims provenance, it is reported in prose rather than the hash-bound tables.

8 Honest Boundary and Reproducibility

8.1 Hosted result: PENDING

At the time of this draft there is no hosted leaderboard score or rank. The hosted playground is a diagnostic service and is labeled as such. The six-set and distribution-drill results are local official-runner measurements. None of them establishes transfer to the hidden evaluation distribution.

There is also no completeness theorem for either branch. Structured countermodels cover chosen families, the finite-model search has carrier and time bounds, and the superposition implementation deepens only until its deadline. Deep-stage outcomes can vary with machine load even though the judge’s verdict on a fixed certificate is deterministic. We make no performance-superiority claim over another solver or prover. The empirical statement is exactly that the archived certificates were accepted in the recorded runs.

8.2 Threats to validity

The public suites influenced solver development. The fourteen v1 residuals motivated particular prover and counter-model changes, and the E168 drill motivated the fixed central groupoid. Full regression shows that those changes integrate without losing earlier rows; it does not measure performance on an independently sampled corpus. The distribution drill is broader but is still published Stage 1 data, and its ground truth may share construction patterns with the public suites.

Wall-clock totals combine Python search, process overhead, and Lean 4 checking under one operator-host environment. They are appropriate for detecting large regressions within that setup but should not be read as portable run-times. Parallel worker load affects time-budgeted deepening, and filesystem caching affects Lean 4 startup. The hosted-playground mean times partially address this environment gap, but playground selection and daily interaction are not the same protocol as final evaluation.

Finally, kernel checking validates each emitted theorem relative to the judge's imported environment and allowed axioms. It does not validate that the benchmark labels, competition problem generator, or provenance narrative are correct. We mitigate the last concern with hashes and exact commands; benchmark construction and judge implementation remain external assumptions.

8.3 Hash-bound evidence

PROVENANCE.json binds the frozen solver's SHA-256 and byte count, the official repository and configuration revisions, each result-ledger hash, the Marathon outputs, the compatibility corpus, and the hosted-playground record. The human-readable claims ledger states the maximum set of supported claims and separately lists prohibited inferences. The repository script `scripts/check_freeze.py` recomputes the solver and ledger hashes, checks accepted-row metadata, verifies the one-file submission layout, and fails closed on drift.

This structure is intended to prevent stale measurements from being attributed to a revised solver. Historical v1 and intermediate v2 artifacts remain named as superseded. The tables in this paper use the current v2.4 bindings from PROVENANCE.json even where an older narrative report contains timings from a previous rerun.

8.4 Replay procedure

Reproduction requires two checkouts: this artifact and the official Stage 2 judge at the pinned revision. In the artifact checkout, first run

```
python3 scripts/check_freeze.py
```

to check immutable bindings. In the official checkout, run `bash scripts/setup.sh`, source `.env.judge`, and invoke the Solo runner with a submission-only directory containing

the frozen solver.py. For example, the public normal set uses

```
python3 -m pipeline.runner \
  --submission /path/to/submission \
  --problems examples/problems/normal.jsonl \
  --output /tmp/normal-results.json
```

The Marathon replay uses `scripts/run_marathon.py`, the canonical `normal_100.jsonl` manifest, and an output directory outside a read-only checkout. Recomputed output hashes should be recorded as a new run rather than overwriting the archived ledgers.

The Lean 4 4.32 compatibility corpus was run in an external directory and is described in the provenance record; it is not presently packaged as a one-command repository test. This is a reproducibility limitation. The official harness, freeze checker, ledger files, and public-runner commands are available locally; hosted playground interaction and any future leaderboard submission require organizer infrastructure.

A measured outer boundary. Shortly before submission the organizers released two additional official sets. On `stage2_stress_test`, announced as mirroring the final leaderboard configuration, the frozen solver accepts 200 of 200 problems with no language-model calls; the set publishes no per-item reference labels, but the per-category verdict counts match the announced composition of twenty-five true and twenty-five false in all four categories, and every accepted verdict is kernel-checked by the judge. On `research_order5_hard` — one hundred research-tier problems in which at least one side is an order-five Austin law, whose false direction provably admits no finite counter-model, and whose ground truth is in part unknown even to the organizers — the frozen solver certifies none within its budgets. Both ledgers are committed and hash-bound. The pair delimits the system precisely: complete coverage of the announced evaluation distribution, and a research frontier, coinciding with open territory in the Equational Theories Project, that finite methods do not reach.

9 Related Work

Equational theories. The Equational Theories Project builds a collaborative implication graph for single magma identities and supplies formal proofs, finite models, and curated problem data [3]. The SAIR task derives its laws and implication setting from this project but imposes a distinct online solver and certificate interface. Cazares analyzes Stage 1 of the challenge and its prediction setting [4]. Stage 2 changes the observable output from a label to a judge-accepted proof or countermodel.

Knuth's study of central groupoids is directly relevant to the E168 witness family [7]. Our use is narrower than the structural theory: we embed one checked non-natural finite model because natural constructions did not separate the

goals encountered in the drill. The infinite-carrier Austin models similarly draw on constructions represented in the Equational Theories Project rather than proposing a general finite-model theorem.

Superposition provers. Vampire [9], E [13], and Prover9 [10] are mature first-order theorem provers implementing variants of resolution, superposition, completion, simplification, and given-clause saturation [9, 10, 13]. Our engine borrows standard ideas—ordered inference, KBO, demodulation, discrimination, tree retrieval, and age/weight selection—but specializes them to unit equations over one binary symbol. The system-level difference is the output contract: the search trace is compiled into a positive Lean 4 equality proof acceptable under the competition policy. We do not compare throughput or coverage against these systems.

Proof assistants and certificate checking. Lean 4 is both the certificate language and the final trusted checker [5]; Mathlib supplies the surrounding library in the official judge environment [14]. The solver follows the proof-producing automation pattern in which a fast untrusted procedure returns an object reconstructed or checked by a small trusted kernel. Finite countermodels are likewise proof objects: exhaustive reflection closes the concrete finite statement, while infinite witnesses use explicit lemmas.

Distillation artifacts. The competition is motivated by work on distilling many-shot context into a compact cheat sheet [6]. The artifact here is executable rather than a natural-language sheet: algebraic families, search control, and proof printers are distilled into one source file. The fixed-configuration language-model measurement is included to document the development path, not to argue that symbolic search and language models are mutually exclusive.

10 Conclusion

The SAIR EQT2 Stage 2 contract rewards an answer only when the same artifact also supplies a machine-checked reason. The solver described here meets that contract with a cheapest-first combination of structured and irregular countermodels, infinite witnesses, and ordered unit superposition, followed by replay through the Lean 4 kernel. Hash-bound local regressions account for all measured public and distribution-drill rows, and a separately labeled hosted playground measurement exercises the announced toolchain. The remaining boundary is explicit: no leaderboard result, hidden-set claim, completeness theorem, or comparative claim is available at this draft stage.

Author contributions (CRediT)

Haobo Ma: conceptualization, methodology, software (solver architecture and the systems foundation), investigation, validation. Wenlin Zhang: conceptualization, methodology, software, investigation, validation, data curation, project administration, writing (original draft). Manuel Israel Cázares: validation (solver freeze verification — hash, file size, and single-file contract compliance; evidence-ledger auditing across the reviewed freezes); methodology (review of experimental design and the claims–evidence boundary); writing (review and editing — full manuscript review and critical review of the presentation and claims).

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